ALM-3

1. Corrosion Mechanisms :-

- Pitting corrosion :- chloride ions in seawater cause the localized holes in the steel.
- Crevice corrosion :- Moisture trapped in joints and cracks create localized attack
- Stress corrosion cracking :- Tensile stress combined with the corrosive marine environment cause cracks.
- Marine environment :- High salt, humidity, oxygen and moisture increase the corrosion rate, while chloride ions damages protective oxide layers

2. Corrosion - prevention methods :-

- Apply protective coatings / paint to isolate steel / iron from seawater and oxygen.
- Use cathodic protection such as sacrificial anodes (or) impressed current systems.

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- Use the corrosion - resistant alloys where practical.
- Ensure proper drainage and regular inspection / and maintainance to prevent water accumilation and detect the damage early.
- Seal the joints / cracks to prevent seawater from entering the causing localized corrosion.